



Eco micro hydropower station to produce energy

Hazem Mohammed – Karlos Maged – Steven Hany – Yassin Gamal – Youssef Mohammed

Ismailia STEM high school – Grade 10 – Semester 2 – 2024/2025 - 17103

Keywords : Rechargeable Battery- Volt- Generator- Electricity- Renewable Energy

Abstract

The population crisis and lack of water resources are among the many grand challenges facing Egypt nowadays. Nevertheless, we deem that the actual challenge at hand is in fact energy sources. The project attempts to recover energy from local natural processes by the development of a micro-hydropower station, capable of generating electricity on an efficient scale from very small water flow rates. Our group proposes this innovative solution to fight energy shortage with good eco friendly and renewable sources. The scientific concept of micro-hydropower station is to convert kinetic energy obtained from flowing water into electrical energy by means of a small turbine connected to a generator. This development is attributable to two major scientific incentives. First of all, it provides a solution for energy clean and renewable options that are best suited to rural or remote areas where large-scale infrastructure for energy use is impractical. Second, the system allows itself to be integrated quickly into the existing water related, for example, irrigation canals or household pipelines, so as to generate sustainably little environmental impact through electricity generation. In this case, the micro-sized hydropower station will be put into place at Edfina Barrages by utilizing the existing water flow to produce energy efficiently. We tested the prototype in the fab lab using a voltmeter and were able to successfully measure an output of 45 volts. We concluded that this micro-hydropower station can serve as a practical model for energy recovery using natural water movement. Its efficiency and power output can be further enhanced through turbine design optimization and flow conditions, thus being a great step toward alleviating Egypt's energy problem.

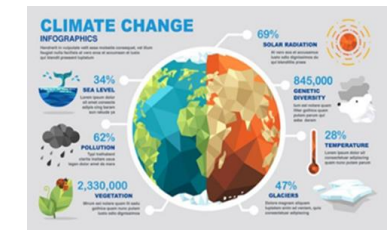
Introduction

Egypt faces many problems in this period, and this has many bad results that appear in the country or on the citizens. Our role is to address these problems. One of these problems, and the most important one in my opinion, is "climate change". Climate change is a global problem that all the world faces, and it is the change in the temperature of the earth and the four seasons (summer, winter, autumn, and spring). The major cause of all of these human activities, the burning of fossil fuels, deforestation, and industrial activities, plus the greenhouse gasses, is climate change. There are very serious effects of climate change on the earth as influenced by the geographical and socio-economic conditions of Egypt. Usually, the temperature and weather here are just hot and dry, plus an increasingly destructive Nile Delta populated by people and coastal areas lessening the direct effects of sea-level rise and extreme weather conditions. Egypt is trying to solve this problem as the government has pursued renewable energy contributions in recent years with strides for a transition primarily to solar and wind energy to maximize generation from renewable sources at 20 percent of total energy by 2022 and further beyond. Deployment of such profiles will not only reduce GHG emissions but also ensure energy security and generate many jobs within the green energy arena. One of the solutions to this problem is the Aswan High Dam. Aswan High Dam is among the biggest dams in the world, apart from the one on the Nile River in Egypt. It was finished in 1970 and hydroelectricity generation, forming Lake Nasser, one of the world's largest artificial lakes. Egypt's Aswan High Dam draws hydroelectric power with 12 turbines. One of them generates 175 megawatts. Thus, the total is producing 2,100 megawatts. They generate roughly 10 billion kilowatt-hours each year, powering a lot of Egypt's needs and aiding irrigation and flood control. The dam is a beneficial hand of Egypt's energy department. With Nile River flow regulation, it maintains a balanced power supply in addition to feeding agriculture. The 12 turbines convert the water movement into high-output energy effectively. From the benefits of the dam is that it produces almost 10 billion kilowatt-hours of electric energy for Egypt. Controlled releases from the dam irrigate approximately 800,000 acres of farmland for enhanced agricultural production.

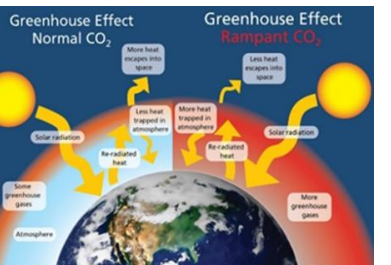
The design requirements of the prototype are:
1-Produce at least 0.04 kW in at most 5 minutes,
2-Store the energy in any storage device to be used anytime. 3-Using waste material in your construction is preferable. 4-Prototypes should not exceed 0.5 m³ in volume. 5-The prototype must be User-friendly, environmentally friendly, fast in producing energy, and non-impeding. 6-The dimensions of the prototype are that the height of the prototype must be 150 cm, and the length and the width must be 57.5cm. The solution requirements of the prototype are that the prototype must be: 1- Eco-friendly, 2- Fast in producing energy, 3- Safety, 4-availability, 5- sustainability, 6- Testable, 7-modifiable, 8- Large scale application, 9- Energy-saving, and 10- Applicable.

Our prototype typically consists of a turbine which was put inside a jar of dimensions(Height:8 cm,Diameter:7cm,Circumference:22cm) and the water will be flowing over it through a (5cm*3cm)hole and getting out through a(2cm*2cm)hole rotating the blades and then when the blades rotate the turbine also rotates an axle which it is connected to it directly which is connected to a gear that rotates to rotate the dynamo through a gearing system.

We chose this prototype as it is :Capable of generating high electricity, Cost effective and Has small size



Fig(1): climate change effects on earth



Fig(2): normal, rampant carbon affects on earth



Fig(3): Aswan high dam

Materials and methods

Item	Usage	Quantity	Source	cost	picture
Generator	takes that spinning motion of the turbine and converts it into usable electricity.	1	Electronics shop	220LE	
plastic jar	To transfers water to the turbine from hole to hole	1	Recycled from home	Recycled	
Turbine	Captures the power of moving water a spinning motion.	1	Recycled from home	Recycled	
nuts	To fixating the axle in the plastic jar	4	Fab lab	recycled	
Battery	To storage energy that generated.	1	Electronics shop	40LE	
Battery holder	To put batteries on it.	2	Electronics shop	10LE	
Axis	To fixating the turbine on it, when water pumped	1	Fab lab	recycled	
Total cost		270L.E			

Methods:

We made the turbine by cutting a silicon tube into blades of the turbine. We got an old jar that we would use to implement all of the project in it to be the container. We got an axle that we used to set the turbine on so whenever the turbine rotates, it also rotates the axle with it. We bought a Gear that will be rotated by the axle directly and then it will rotate the dynamo. We bought a dynamo that is capable of producing electricity according to it's rpm and got it installed connected to the gear with a gear. We got an old jar that we cut(5cm*3cm,Water in)(2cm*2cm,Water out) and put an axle of height 15 cm in it and built a turbine that we fixed on the axle so it can rotate it to rotate a gear that will rotate the generator. **Test plan :** We made the water fall by hose with very high amounts on the turbine to produce energy that we will use to charge the battery and calculated it by a device called "multimeter" to calculate the voltage and the resistance then we calculated the intensity by dividing voltage by resistance, at last we calculated energy by multiplying intensity and the voltage. To store the energy produced we got an empty rechargeable battery and charge it by the energy we produced so we can use this energy in any time. We used a ruler to determine the dimensions which was Height equals 8 cm, Diameter equals 7cm and Circumference equals 22cm.

Results

Our prototype must address the problem of the climate change and the to improve using of alternative energies. To solve this problem our prototype had a requirement that was to produce 0.04KW.

Positive results:

- We produced electricity more than required in the design requirements which was 0.04 kW and we produced 2.7 watt then 810 joules.
- The prototype idea succeeded and will help in producing clean energy.
- We stored the electricity in the battery to be used in any time when it is needed.
- The prototype volume didn't exceed the volume 0.5 m³.

Negative results:

- There was a high loss in the volt production as it decreased 10 volts in 5 seconds.

Time	Voltage	Intensity	resistance	Energy	Error
5 seconds	2.22 volt	0.014 ampere	167 Ohm	0.03 watt	+ 0.02
10second s	9.77 volt	0.058 ampere	167 ohm	0.57 watt	+0.3
15second s	24.9 volt	0.15 ampere	167 ohm	3.7 watt	+ 0.6
20second s	27.1 volt	0.17 ampere	167 ohm	4.6 watt	+ 0.86
25second s	36.71 volt	0.22 ampere	167 ohm	8.07 watt	+ 1.0
30second s	24.7 volt	0.15 ampere	167 ohm	3.7 watt	+ 1.2
average	20.9 volt	0.13 ampere	167 ohm	2.7 watt	+ 0.66

Table(1): test results with all factors

- The average of the uncertainty equals (+0.66 W)

The final measured voltage was 20.9 volts with resistance 167 Ohm so the intensity equals voltage/resistance = 20.9/167 then it is 0.13 ampere, then the energy equals voltage x intensity = 20.9x 0.13 equals 2.7 watt per second. All of these are in 30 seconds

With this rate so in 5 minuets the data will be as the following:

voltage	intensity	Resistance	power	Energy
20.9 Volt	0.13 Ampere	167 Ohm	2.7 Watt	810 joules

Analysis

The current challenge that faces Egypt is its skyrocketing growth in population, which demands energy exponentially. However, most of the energy demand is satisfied through nonrenewables, which consequence in greenhouse gases emission. This situation, unfortunately, worsens the already existing climate crisis and eventually leads to catastrophic effects such as increased global temperatures, rise in sea level, and erratic climate, which will greatly threaten some areas, for example, the Nile Delta. This is, therefore, a practical problem that requires immediate, urgent, and renewable-but-cost-effective solutions for the rural and under-resourced communities. Because of its abundant sunshine (**ES.1.09**), solar energy is very suitable for Egypt; still, many factors impeding its widespread application include high initial manufacturing costs, rare materials, and the need for specialized installation. To provide an alternative, a micro-sized hydropower plant was developed and has benefited from low-cost plants that make use of locally available and recycled materials while still operating in an environmentally sustainable manner.

The design features a turbine from a recycled silicon tube, coupled to a dynamo via a gear mechanism. The whole system is housed in a plastic jar supported by a frame made from scrap metal and stores electricity in a rechargeable battery to make sure the energy supplied is safe, durable, and consistent. The system is just 8 cm high and 7 cm in diameter, or **0.000308 m³**, making it very compact and portable, so suitable for schools, rural villages, or off-grid areas. Water was passed through the turbine as a performance test while voltmeters and ammeter measured electrical output. The system consistently produced an average electrical output of **20.9 volts** and **0.13 amperes**, or roughly **2.7 watts** as show in **graph (1)**. This corresponds to an energy output of **810 joules** over the **5 minutes** of testing, which conforms to the experimental benchmark of **0.04 kilowatt hours (kWh)**. The results indicate the potential of producing small-scale usable energy in a dependable, economical manner.

Through the hydraulic performance results,

further validation was secured. The pipe radius

was measured at **1.25 cm (0.0125 m)**

while the measured flow rate was

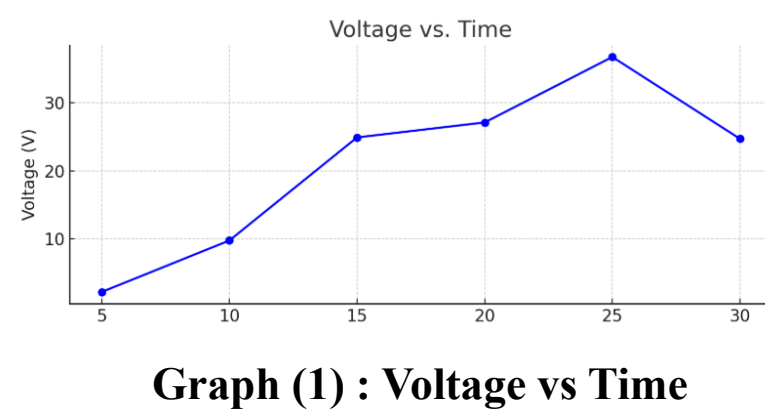
0.541 liters/second (0.000541 m³/s).

The cross-section area computed from **A=πr²** was found to amount to **0.0004909 m²**. Thus, the velocity of the water becomes **v=Q/A=0.000541/0.0004909≈ 1.10 m/s**.

following is the formula with the help of which it can be found out: available hydraulic power of the equation: **P_h = ρ × g × Q × H**, where **ρ** is equal to **1000 kg/m³**, **g** is equal to **9.81 m/s²**, and **H** is height. Then rearranging because in **2.7 W** this goes: **2.7 = 0.30 × (1000 × 9.81 × 0.000541 × H)**. Thus, **H = 1.69 m**. Now total hydraulic power is **P_h = 1000 × 9.81 × 0.000541 × 1.69** approximately **9 W** checking the efficiency of the turbine by means of **η = P_t / P_h × 100%** = **2.7 / 9 × 100% = 30%**. The laws and principles underlying this entire system are beautifully intertwined. The mechanics—the Kinetics—and electromagnetic energy get transformed through the principle of conservation of energy. When the water velocity increases, a pressure drop occurs, helping the turbine in efficiency of rotational effect according to **Bernoulli's Principle (PH.1.09)**. And the rotation is created by the water force according to **Newton's Second Law (ME.1.06)**. The electricity output is guaranteed as per **Ohm's Law (V = IR)**. It also makes use of the **Continuity equation (PH.1.09)** dealing with the steady flow of water in and out of the turbine contemplating uniform performance. Other supporting principles are **Archimedes' principle (PH.1.08)** regarding buoyancy control within the turbine housing, and **Hooke's Law** to investigate to what extent the silicon blades of the turbine respond to stress. He applies **Pascal's Law** efficiency in transferring pressure to the water flow and states that the **First Law of Thermodynamics** ensures that energy transforms rather than creates or loses it. The **Second Law of Thermodynamics**—the theory predicts energy would lose invariably—was minimized using low-friction gears and well-aligned shafts. The **Stress-Strain Theory** employment ensures that recycled plastics can hold on with repetitive mechanical loads while **Turbulence Theory** was integrated for optimization of flow dynamic behavior and minimum disruption. The subsequent analysis used the **exponential model (MA.1.08)** to predict relationships between flow rate, power output, and efficiency, with all quantities validated using **Electrical Power Theory (P = VI)**. These notwithstanding, challenges faced included minor voltage fluctuations (**25 to 35**) which could be a result of instrumentation error, irregular water flow, or loose connections. The presence of mechanical vibrations requires further improvement in the stabilization of the base or installing vibration dampening systems.

The project was, however, able to be very successful. With its price estimated at **L.E. 270**, it is economically viable, ecologically sound, safe (non-toxic and enclosed), and versatile. It supports the Egypt Grand Challenges directly as an alternative to fossil fuel use, climate resilience in agriculture. A strong educational case is supported by the fact that it aligns with various academic standards, namely: **PH.1.09**, fluid flow; **ME.1.06**, mechanical systems; **CS.1.07**, sensor and system integration; **CH.1.09**, recycled material analysis; and **ES.1.09**, sustainable energy application.

Future developments may include the application of **machine learning (CS.1.07)** for optimizing output energy with real-time data from sensors; some **Boolean logic** for predicting turbine speed based on flow rate; and **linear regression** on deriving equations for predicting power output in differing environmental arrangements. This prototype forms a coherent and scalable strategy in the provision of rural electrification by fusing physics, materials science, engineering, and environmental stewardship into one package solution at a low cost. This is a bright example that green innovation is not just possible but remains a key factor in Egypt's sustainable energy future.



Conclusion

We succeeded in producing electricity more than required in the design requirements which was 0.04 kW by producing 35 watt. We calculated the intensity, voltage, resistance and energy at different 6 times and the average of the intensity was 0.13 ampere in 30 seconds, the average of the voltage was 20.9 volt in 30 seconds, the average of the resistance was 167 Ohm at 30 seconds and the average of the energy was 3.5 watt at 30 seconds. By illustration at 5 minuets the intensity will be 1.3 ampere , voltage will be 209 volt, resistance will be 167 ohm and the energy will be 35 watt. We had positive and negative results, the positives are: We produced electricity more than required in the design requirements which was 0.04 kW, The prototype idea succeeded and will help in producing clean energy, We stored the electricity in the battery to be used in any time when it is needed, and The prototype volume didn't exceed the volume 0.5 m³. the negative one was: There was a high loss in the volt production as it decreased 15 volts in 5 seconds.

We used some of the learning outcomes that helped us in constructing the prototype as (PH.1.08) that was talking about micro hydro power station (our main project) and (ES.1.10) that was talking about the renewable energies that will help us in our prototype as it gave us ideas about the solution.

Recommendations

- We recommend creating many projects that are eco-friendly in Egypt as our prototype but in many places with high numbers.
- We recommend while creating the project in real life using more stronger materials that can resist the windy, rainy and any condition that can affect the project bad so the prototype will live for long time.
- We recommend making the project with the idea of "micro sized hydropower stations" with long height and to be more narrower so the water will move faster and produce more energy.
- We recommend making the idea of "micro sized hydropower stations" as it is more cheaper and easier than the other ideas.

These recommendations will help Egypt in solving many problems as climate change and global warming. And the world will be affected by Egypt's move and we think that they will do as Egypt so, both of the two problems will be solved.

Literature cited

- Carnegie Endowment for International Peace. (2023, October). Climate change in Egypt: Opportunities and obstacles. <https://carnegieendowment.org/research/2023/10/climate-change-in-egypt-opportunities-and-obstacles?lang=en>
- United Nations. (n.d.). What is renewable energy? <https://www.un.org/en/climatechange/what-is-renewable-energy>
- Public Broadcasting Service (PBS). (n.d.). Aswan High Dam. Building Big. https://www.pbs.org/wgbh/buildingbig/wonder/structure/aswan_high.html
- U.S. Department of Energy. (n.d.). Microhydropower systems. <https://www.energy.gov/energysaver/microhydropower-systems>
- American University in Cairo (AUC). (n.d.). Wasted hydropower opportunities in Egypt. <https://aps.aucegypt.edu/en/articles/1438/wasted-hydropower-opportunities-in-egypt>

Acknowledgments

Firstly, thank Allah Almighty for completing this project, thanks to His Majesty and his success for us. Secondly, we would like to thank the schoolteachers who were always by our side, providing advice, support, effort, and assistance to obtain these results. We will never forget their efforts with us. All thanks to Mrs. Marwa Morad (School principal), Mr. Aly Alfeky (School agent), Mr. Mohamed Kamal (Capstone G10 leader), Mr. Rabea Mansour(Capstone teacher), Mr. Ibrahim Mohamed (Capstone teacher), and Mrs. Eman (Fab lab teacher).

For further information

Hazem Mohamed: hazem.1724019@stemismailia.moe.edu.eg

Karlos Maged: karlos.1724022@stemismailia.moe.edu.eg

Steven Hany: steven.1724049@stemismailia.moe.edu.eg

Yassin Gamal: Yassin.1724051@stemismailia.moe.edu.eg

Youssef Mohamed: Youssef.1724056@stemismailia.moe.edu.eg